

THE GRYP

Project Brief for the Development Team

What is GRYP?

GRYP is a **white-labeled community and member engagement platform built specifically for credit unions**. Think of it as a private, branded version of Discord or Slack, but designed for credit union members instead of gamers or office workers.

A credit union signs up with GRYP. Their members get a mobile app (iOS and Android) and web access that is branded to look like the credit union's own platform. Inside, members can chat, join channels, attend events, engage with content, message each other, and interact with local businesses (stakeholders/vendors). The credit union's admin team manages everything through an admin dashboard.

The key insight: credit unions are built on community, but they have zero digital community infrastructure. Members only interact with their credit union to check balances or apply for loans. GRYP gives them a reason to engage every day.

How Credit Unions Are Involved

Credit unions are the **customers**. They are the ones paying for GRYP. Each credit union gets their own isolated instance of the platform with their own branding, their own members, and their own data.

The relationship works like this:

- **James (the client)** is building GRYP as a company that sells to credit unions
- Each credit union that signs up becomes a **tenant** on the platform
- The credit union's staff become **CU Admins** who manage their community
- The credit union's members become **users** of the app
- Local businesses that want to reach CU members become **stakeholders/vendors**

James does not sell to individual people. He sells to credit union institutions. One sale to a credit union with 50,000 members means 50,000 users overnight.

User Roles in the System

Role	Who They Are	What They Do
Super Admin	James / GRYD team	Manages the entire platform, creates tenants, oversees all credit unions
CU Admin	Credit union staff	Manages their credit union's community: channels, members, content moderation, events, settings
Stakeholder / Vendor	Local businesses	Posts products/services, engages with CU members, participates in marketplace
Member	Credit union members	Joins channels, chats, attends events, browses marketplace, earns badges

Platform Architecture

Layer	Technology	Purpose
Mobile App	React Native (Expo)	iOS and Android app for members, CU admins, and stakeholders
Web App	Expo Web (deployed on Vercel)	Same app accessible via browser at thegryd.io
Backend API	Node.js / Express	REST API gateway that handles all business logic
Database	MongoDB	Stores users, messages, channels, events, posts, settings
Real-time	WebSocket (Socket.IO)	Live messaging, typing indicators, presence (online/offline)
Media	Cloudinary	Image and file uploads
Voice/Video	Agora	Voice channels and video calls
Push Notifications	Expo Notifications	Message alerts, call notifications, event reminders
Auth	JWT tokens	Login, signup, OTP verification, session management

Every Feature and Page in the App

Member Dashboard (`app/(main)/index.tsx`)

The main screen members see after logging in. Contains:

- Channel list for the credit union's community (text channels, announcement channels)
- Content feed with posts from admins, stakeholders, and members
- Real-time messaging in channels with text, images, files, voice messages
- Like, comment, and reshare on posts
- Pull-to-refresh for new content
- Delivery status icons on sent messages (sending, sent, failed with tap-to-retry)
- Typing indicators when others are typing
- Infinite scroll for message history

CU Admin Dashboard (`components/CreditUnionAdminScreen.tsx`)

The main screen for credit union administrators. This is the largest and most complex screen in the app (11,000+ lines). Contains:

- **Channel Management:** Create, edit, delete text/announcement/voice channels. Organize into categories.
- **Member Management:** View all members, assign roles, ban/mute/suspend members, manage permissions
- **Content Feed:** Post announcements, articles, media. Pin important messages.
- **Content Moderation:** Automated word filters, manual review queue, prohibited words list, severity levels
- **Events:** Create and manage events and announcements with dates, locations, descriptions
- **Server Settings:** Change community name, logo, banner, description
- **Invite System:** Generate invite links and codes for new members
- **Stakeholder Management:** Invite vendors, assign badges (Gold/Silver/Bronze), manage their access
- **Custom Roles:** Create roles with specific permissions (can post, can moderate, can invite, etc.)
- **Sub-communities (Subgrids):** Create separate communities within the main CU community
- **Voice Channels:** Host live voice/video calls with mute, camera, and speaker controls

- **Privacy Settings:** Control member visibility, profile privacy, data sharing
- **Notification Settings:** Configure push notification preferences
- **Engagement Settings:** Leaderboard configuration, points system, badges
- **Moderation Dashboard:** Review flagged content, take action (warn, mute, ban)
- **Theme Toggle:** Light/dark mode

Direct Messages (app/(main)/direct-messages/index.tsx + components/DirectMessagesScreen.tsx)

- One-on-one private messaging between any two users
- Text, images, files, voice messages
- Delivery status and typing indicators
- Online/offline presence indicators
- Conversation list with last message preview

Sub-Channel View (app/(main)/sub-channel.tsx)

- Dedicated view for individual channels within a subgrid
- Full messaging with all media types
- Message pinning and moderation controls
- Voice message recording and playback

Profile Page (app/(main)/profile.tsx)

- User profile with avatar, name, bio
- Edit profile information
- View engagement stats and badges

Top Contributors / Leaderboard (app/(main)/top-contributors.tsx)

- Ranked list of most active community members
- Points breakdown by activity type
- Badge showcase

Privacy Settings (app/(main)/privacy-settings.tsx)

- Control who can see your profile
- Control who can send you direct messages

- Visibility toggles for online status, activity, etc.

Authentication Pages

- **Login:** Email/phone + OTP verification
- **Signup Member:** Join a credit union community with invite code
- **Signup Stakeholder:** Register as a vendor/business
- **Super Admin Signup:** Protected registration for platform administrators
- **Join via Link:** Accept invite link to join a specific community (app/join/[token].tsx)

Incoming Calls (components/IncomingCallOverlay.tsx)

- Full-screen overlay when someone calls you
- Ringtone audio playback
- Accept/decline buttons
- Caller information display

Key Backend Endpoints

Area	What It Handles
Auth Routes	Login, signup (member, stakeholder, super admin), OTP verify, password change, token refresh, logout
Community Routes	CRUD for subgrids, channels, posts, messages, events, categories, members, roles, invites, moderation, privacy
WebSocket Service	Real-time messaging, typing indicators, presence tracking, call signaling

Key Middleware

- **Auth Middleware:** JWT token verification on all protected routes
- **Subgrid Access:** Ensures users can only access communities they belong to
- **Privacy Middleware:** Enforces member visibility and privacy settings
- **Rate Limiting:** Protects against brute force on signup/login endpoints

What is Being Built Next

Feature	Status	Description
Marketplace	Planned	Stakeholders list products, members purchase. Stripe Connect for card payments, crypto wallet (USDC on Polygon) as optional payment
Embed SDK	Planned	API/SDK so credit unions can embed GRYD community features directly into their existing mobile banking app
AI Insights	Future	Engagement recommendations, member sentiment analysis, predictive retention, benchmarking across CUs
Glassmorphism UI	In Progress	iOS 26-style frosted glass design for CU Admin dashboard (currently testing)

Business Model (How James Makes Money)

- **Monthly retainer per credit union** (not per user). Flat fee based on CU size.
- **No per-seat pricing.** If 50,000 members join, the price stays the same.
- **Managed service included:** Onboarding, training, monthly optimization reports, ongoing support.
- **Future revenue:** Marketplace transaction fees, premium analytics, custom integrations.
- **Positioning:** GRYD is sold as a "member engagement solution," not as software. James is not selling a tool. He is selling the outcome of engaged, loyal members.

Quick Reference: File Structure

Path	What It Contains
frontend/ios-app-new/	The entire React Native (Expo) app: mobile + web
frontend/ios-app-new/app/	All screens/pages (Expo Router file-based routing)
frontend/ios-app-new/components/	Reusable components (CU Admin, DM screen, overlays, skeletons)
frontend/ios-app-new/contexts/	React contexts (WebSocket, Notifications)

frontend/ios-app-new/hooks/	React Query hooks for all API calls
frontend/ios-app-new/lib/	Utilities (API client, theme, haptics, message queue)
backend/api-gateway/	Node.js/Express backend
backend/api-gateway/src/controllers/	Business logic (auth, community, billing)
backend/api-gateway/src/middleware/	Auth, privacy, rate limiting, access control
backend/api-gateway/src/routes/	API route definitions
backend/api-gateway/src/services/	WebSocket service

Prepared for the GRYD development team. Confidential.